EE4-07
EE9-CS7-23
EE9-SO11

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE/EIE PART IV: MEng and ACGI

# CODING THEORY

Tuesday, 28 April 10:00 am

Time allowed: 3:00 hours

**There are THREE questions on this paper.**

**Answer ALL questions.**

*All the questions carry equal marks.*

**Any special instructions for invigilators and information for candidates are on page 1.**

Examiners responsible First Marker(s) : W. Dai
Second Marker(s) : C. Ling

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# EE4-07 Coding Theory

## Instructions for Candidates

Answer all three questions. Each question carries 25 marks.

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1. (The Elementary)

(a) Let $\mathcal{C} \subset \mathbb{F}_2^7$ be a linear code. Its parity check matrix is given by

$$\boldsymbol{H}' = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}.$$

i Use Gaussian elimination to change the parity matrix into the form of $\boldsymbol{H} = [\boldsymbol{I}\ \boldsymbol{A}] \in \mathbb{F}_2^{3\times 7}$ where $\boldsymbol{I}$ is the identity matrix. [2]

ii Find the corresponding generator matrix $\boldsymbol{G} \in \mathbb{F}_2^{4\times 7}$ in the form $[\boldsymbol{B}\ \boldsymbol{I}]$. [2]

iii Assume that a message $\boldsymbol{m}_1$ is encoded into a codeword $\boldsymbol{c}_1$ using $\boldsymbol{G}$. The codeword $\boldsymbol{c}_1$ is transmitted over an binary symmetric channel. Let the received word be $\boldsymbol{y}_1 = [0\ 0\ 1\ 1\ 1\ 1\ 0]$. Compute the syndrome vector $\boldsymbol{s}_1$. Find the output of the minimum (Hamming) distance decoding, say $\hat{\boldsymbol{c}}_1$, and the corresponding transmitted message $\hat{\boldsymbol{m}}_1$. [3]

iv Assume that a message $\boldsymbol{m}_2$ is encoded into a codeword $\boldsymbol{c}_2$ using $\boldsymbol{G}$ The codeword $\boldsymbol{c}_2$ is transmitted over an binary erasure channel. Let the received word be $\boldsymbol{y}_2 = [1\ 0\ 1\ 1\ 0\ ?\ ?\ 0]$. Set the question marks in $\boldsymbol{y}_2$ to zero and compute the corresponding syndrome vector $\boldsymbol{s}_2$. Find the transmitted codeword $\boldsymbol{c}_2$ and the message $\boldsymbol{m}_2$. [3]

(b) Let $\mathcal{C}$ be a linear code. Its distance can be defined as

$$d(\mathcal{C}) = \min_{\boldsymbol{c}\in\mathcal{C}:\ \boldsymbol{c}\neq\boldsymbol{0}} \mathrm{wt}(\boldsymbol{c}).$$

i State the relationship between the distance $d(\mathcal{C})$ and the columns of its parity check matrix $\boldsymbol{H}$. Prove your answer. [3]

ii What is the distance of the binary linear code defined in Question 1.(a)? Justify your answer. [2]

(c) i Let $a(x) = x^2 + 1 \in \mathbb{F}_2[x]$ and $b(x) = x^3 + x^2 + 1 \in \mathbb{F}_2[x]$. Find $f(x) = \gcd(a(x), b(x))$. [2]

ii Apply Bï¿œzout's identity to compute $g(x) \in \mathbb{F}_2[x]$ and $h(x) \in \mathbb{F}_2[x]$ such that $f(x) = g(x)a(x) + h(x)b(x)$. [2]

(d) Consider the finite field $\mathbb{F}_{16}$ detailed below

| | $\mathbb{F}_2[z]/z^4+z+1$ | | $\mathbb{F}_2[z]/z^4+z+1$ |
|---|---|---|---|
| 0 | 0 | $z^7$ | $z^3+z+1$ |
| 1 | 1 | $z^8$ | $z^2+1$ |
| $z$ | $z$ | $z^9$ | $z^3+z$ |
| $z^2$ | $z^2$ | $z^{10}$ | $z^2+z+1$ |
| $z^3$ | $z^3$ | $z^{11}$ | $z^3+z^2+z$ |
| $z^4$ | $z+1$ | $z^{12}$ | $z^3+z^2+z+1$ |
| $z^5$ | $z^2+z$ | $z^{13}$ | $z^3+z^2+1$ |
| $z^6$ | $z^3+z^2$ | $z^{14}$ | $z^3+1$ |

i Compute $\beta_1 = (z+1)\cdot(z^3+z^2+z+1) \in \mathbb{F}_{16}$. [2]

ii Compute $\beta_2 = (z^3+z^2+1)^{-1}\cdot(z^2+1) \in \mathbb{F}_{16}$. [2]

iii Compute the polynomial $f(x) = (x-(z^2+z))\cdot(x-(z^2+z+1)) \in \mathbb{F}_{16}[x]$. [2]

2. (Prime Numbers)

(a) i Let $p$ be a prime number, $a > 1$ and $b > 1$ be two integers. Prove that $p|(ab)$ implies that $p$ divides one of the numbers $a$ and $b$, or equivalently, if $p \nmid a$ then $p|b$. [3]

ii Let $p$ be a prime number, and $a_i > 1$, $i = 1, \cdots, n$, be integers. Use mathematical induction to prove the statement that $p|(a_1 a_2 \cdots a_n)$ implies that $p|a_i$ for some $i \in \{1, 2, \cdots, n\}$. [3]

iii Find an example where $a > 1,\ b > 1,\ c > 1$ are integers, $c|(ab)$ but $c \nmid a$ and $c \nmid b$. [2]

(b) Let $a, b \in \mathbb{Z}^+$.

i Show that if $\gcd(a, b) = 1$, then the multiplicative inverse $a^{-1} \pmod{b}$ exists. [3]

ii Prove that if the multiplicative inverse $a^{-1} \pmod{b}$ exists, then $\gcd(a, b) = 1$. [3]

Hence, it can be concluded that the multiplicative inverse $a^{-1} \pmod{b}$ exists if and only if $\gcd(a, b) = 1$.

(c) Let $p$ be a large prime number. Let $g$ be a primitive element in the finite field $\mathbb{F}_p$.

i What is the computational complexity to compute $g^a \in \mathbb{F}_p$ in general, where $a \in [1, p-1]$ is a large integer? Explain the typical computation method to achieve this computational complexity. [3]

ii What is the computational complexity of compute the discrete logarithmic function $\log_g(b) \in [1, p-1]$ for a given $b \in \mathbb{F}_p^*$? Explain the typical computation method. [2]

iii Consider the following key exchange problem. Alice and Bob want to share a secret key but their information exchange could be observed by their adversary Eve. The goal for Alice and Bob is to share a key without making it known to Eve.
One solution is the famous Diffie-Hellman key exchange scheme. Suppose that Alice picks a secrete integer $a$ and Bob's secrete integer is $b$. Describe the rest steps of Diffie-Hellman key exchange scheme. [3]

iv Consider ElGamal public key cryptography scheme. Let $a' \in [1, p-1]$ is the private key for Alice. How would Alice construct her public key?

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How would Bob encrypt his message $m$ using Alice's public key? And how can Alice decrypt the message from Bob? [3]

3. (Codes)

(a) i Consider a code $\mathcal{C} \in \mathbb{F}_q^n$ where $n$ is the codeword length. Let $M = |\mathcal{C}|$ be the number of the codewords and $d$ be its distance. Hamming (or sphere-packing) bound provides an upper bound on the number of codewords $M$. State Hamming bound and prove it. [4]
A code that attains Hamming is called a perfect code.

ii Consider a *linear* code $\mathcal{C} \in \mathbb{F}_q^n$ with parameters $[n, k, d]$. State the corresponding Singleton bound and prove it. [2]
A linear code that attains Singleton bound is called a maximum distance separable (MDS) code.

(b) i Let $r > 2$. A binary Hamming code $\mathcal{H}_r$ is defined as the code with a parity check matrix $\boldsymbol{H}_r$ whose columns consist all the nonzero column vectors of $\mathbb{F}_2^r$. Specify the parameters $[n, k, d]$ of the binary Hamming code $\mathcal{H}_r$. [3]

ii State whether a binary Hamming code $\mathcal{H}_r$ is a perfect code or not. Prove your answer. [3]

iii State whether a binary Hamming code $\mathcal{H}_r$ is an MDS code or not. Prove your answer. [3]

(c) Define the generating function of a codeword $\boldsymbol{c} = [c_0, \cdots, c_{n-1}] \in \mathbb{F}_q^n$ as $c(x) = \sum_{i=0}^{n-1} c_i x^i \in \mathbb{F}_q[x]$. A *linear* code $\mathcal{C}$ is called a cyclic code if for all $c(x) \in \mathcal{C}$, it holds that $u(x)\, c(x) \mod x^n - 1 \in \mathcal{C}$ for an arbitrary $u(x) \in \mathbb{F}_q[x]$. For a cyclic code $\mathcal{C}$, choose its generator polynomial $g(x) \in \mathcal{C}$ as the nonzero monic polynomial of least degree.

i Prove that for all $c(x) \in \mathcal{C}$, it holds that $c(x) = u(x)g(x)$ for some $u(x)$. [2]

ii Construct a BCH code on $\mathbb{F}_2$ with $n = 31$ and $d \geq 7$ by finding an appropriate generator polynomial $g(x)$ in the following way.
Find the corresponding cyclotomic cosets $C_i$, $i = 0, 1, \cdots, 8$. [3]
Let $\alpha$ be a primitive element in $\mathbb{F}_{32}$. Find a generator polynomial $g(x)$ to guarantee $d \geq 7$. Write it using minimal polynomials. [2]

iii Let $\alpha$ be a primitive element of $\mathbb{F}_q$. Construct a cyclic code $\mathcal{C}[n, k] \in \mathbb{F}_q^n$ where $n = q - 1$ by using the generator polynomial

$$g(x) = (x - \alpha)(x - \alpha^2) \cdots (x - \alpha^{n-k}).$$

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What can you say about its distance? Justify your answer. [3]